Examiner only

4. A beam of monochromatic light is shone normally (at right angles) on to a diffraction grating.

(a) **Explain** in clear steps why bright beams emerge from the grating at angles, θ, to the normal given by the equation:

$$d \sin \theta = n\lambda$$

[3]

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(b) The angles, θ, at which the bright beams emerge are given in the table below.

Order, n	**θ (mean)**	
0	0	
1	16	
2	35	
3	58	

(i) Plot a graph of sin θ (y-axis) against n (x-axis) on the grid provided. [3]

(ii) **Use your graph** to determine the wavelength of the light. The separation between the centres of slits in the grating is 1 800 nm. Show your working. [3]

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